

REVISIONS #7



PURE MATHEMATICS

ANSWER ALL QUESTIONS

TIME: 60 MINUTES

PASS MARK: 35/50

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- 1 Solve the following inequalities
 - a) $|x + 6| \leq 3|x - 2|$ [3]
 - b) $|3^{-x} - 54| < 27$ [4]
- 2 a) Express $\frac{1}{x^2(x-3)}$ in the form $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-3}$ [3]
b) Prove the identity $\sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = \sec \theta - \tan \theta$ [4]
- 3 The variable V varies jointly as the square root of r and inversely as the square of t ,
When $r = 16$; $t = 3$; $v = 8$.
 - a) Express V in terms of r and t [3]
 - b) Find the value of t when $V = 10$ and $r = 100$ [2]
- 4 a) Find the length of the line joining the origin to the point of intersection of the
line $y = x$ and the curve $y = \frac{1}{x^2}$ [5]
b) Calculate the radius of the circle which passes through the points $A(0; 0)$, $B(2; 10)$
and $C(10; -2)$ [5]
- 5 The function $f(x) = 2 - \frac{1}{x}$, $x > 0$
 - a) Sketch the graph of $f(x)$ and state its range. [3]
 - b) Find $f^{-1}(x)$ and state its domain [3]
 - c) Calculate the value of x for which $f(x) = f^{-1}(x)$ [3]
- 6 Given that $x^2 - 4$ is a quadratic factor of $x^3 + ax^2 + bx + 4$, find the values of
 a and b . Hence factorise the polynomial completely. [6]
- 7 Using any method of your choice, solve the following simultaneous equations
$$\begin{aligned}x + y + z &= -2 \\x + y - z &= 2 \\x + 2y - 3z &= -3\end{aligned}$$
 [6]

Answers are available at totoacademy.org.zw

THE END

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